

Lecture 1: Course organisation
Lecture 2: Scientific Discovery: Understanding from Data
From AI to Deep Learning

Deep Learning for Scientific Discovery

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Course Organisation



About Me

Institute of Theoretical Physics, UJ
deep learning for scientific problems & quantum machine learning

Career path:

- 2010–14 PhD, Jagiellonian U. (Bose–Einstein condensates, quantum optics)
- 2015–17 TU Eindhoven (Rydberg quantum simulators)
- 2017–21 PAS Warsaw (topological matter, metrology, deep learning)
- 2021–26 ICFO Barcelona (quantum simulators & information, deep learning)
- 2025–26 Qilimanjaro Quantum Tech (quantum machine learning)
- 2026– back at UJ

Taught deep learning to physicists in Barcelona (Master's in Quantum Science & Technologies).

This course is new: for the first time the audience is *not* only physicists. Your feedback will shape it.

Getting the Most from This Course

Bring your own research:

- think of a problem from *your* field while we learn each method
- ask how the idea would apply to *your* data
- the final project is built on *your* research question

Learn from each other:

- physicists, chemists, mathematicians, biologists, geographers...
- the same method often solves problems in very different fields

The key question

What does *your* data look like?

- images & maps
- sequences & time series
- molecules, networks, graphs
- fields on a grid (simulations, climate)
- tables of measurements
- equations, proofs, no data at all...

Keep your answer in mind — we will come back to it today.

The Aim of This Course

Every scientific revolution arrives with a **new instrument**:

the telescope · the microscope · X-ray crystallography · the particle accelerator · the computer

Machine learning is the instrument of our era.

This course is *not*

a recipe book of libraries and buttons to press.

This course *is*

how deep learning works, *why* it works, *where* it fails — and how to use it on *your* research problem.

Tentative Roadmap

I. Foundations

1. From AI to deep learning
2. Neural networks: what they can represent
3. How neural networks learn
4. Training in practice
5. Convolutional networks & symmetry
6. Unsupervised learning & autoencoders
7. Sequences, attention, Transformers
   biology: AlphaFold

II. Behind the breakthroughs

8. Generative models
   materials & chemistry
9. Reinforcement learning
   mathematics: AlphaTensor
10. Learning with known rules
    physics & weather
11. Interpretability & symbolic regression
    reading laws out of networks
12. ML $\leftrightarrow$ quantum computing

III. Frontiers

13. Safety & security of neural networks
14. Large language models I
15. LLMs II & the limits of AI

This roadmap is *tentative* — topics, order and examples will be adjusted as needed.

Laboratories: Google Colab



- **Python** + NumPy, then PyTorch or JAX
- runs in the browser — *no installation*, free GPUs
- lab k mirrors lecture k
- notebooks shared before each lab

To do before Lab 1:

- a Google account → open colab.research.google.com
- new to Python? → dedicated *warm-up notebooks* will be provided

Assessment

1. Project — a short scientific article

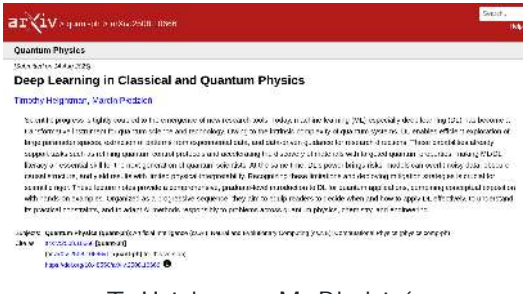
- topic of *your choice* (e.g. from your research) or one of the topics I will propose
- **full code** — the complete, reproducible pipeline
- **manuscript** — written as a short article, with proper presentation of methods and results

2. Oral exam

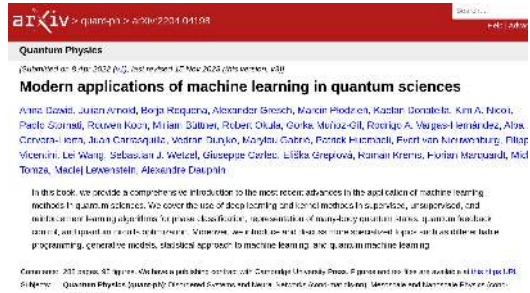
- discussion of your project: *methods* and *results*
- plus **two questions** on deep learning (lecture material)

Start thinking about your project now — the labs will help you build it step by step.

Main Reading: the Two Course Books



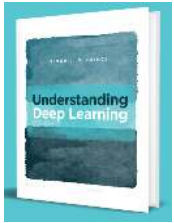
T. Heightman, M. Płodzień,
Deep Learning in Classical and Quantum Physics, Springer (to appear, 2027)
arXiv:2508.10666

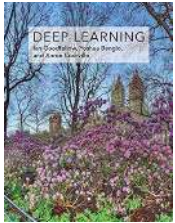


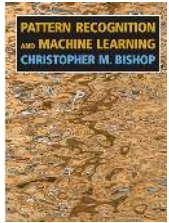
A. Dawid, J. Arnold, . . . , M. Płodzień *et al.*,
Machine Learning in Quantum Sciences, Cambridge UP (2025)
arXiv:2204.04198

Read along: the lectures follow these two books — both free on arXiv.

Further Reading: Deep Learning Classics

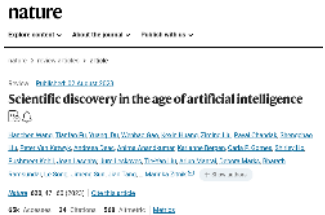






- S. J. D. Prince, *Understanding Deep Learning*, MIT Press (2023) — free online
- I. Goodfellow, Y. Bengio, A. Courville, *Deep Learning*, MIT Press (2016)
- C. M. Bishop, *Pattern Recognition and Machine Learning*, Springer (2006)
- R. S. Sutton, A. G. Barto, *Reinforcement Learning*, 2nd ed., MIT Press (2018)

Literature: Machine Learning for Science


nature

Scientific discovery in the age of artificial intelligence

H. Wang *et al.*, Nature **620**, 47 (2023)


nature reviews physics

On scientific understanding with artificial intelligence

M. Krenn *et al.*, Nat. Rev. Phys. **4**, 761 (2022)

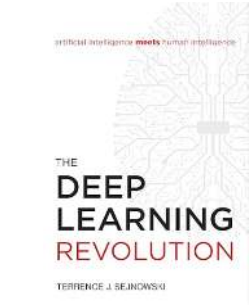

A high-bias, low-variance introduction to Machine Learning for physicists

P. Mehta *et al.*, Phys. Rep. **810**, 1 (2019)

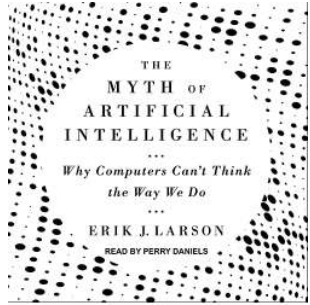

Introduction to Machine Learning for the Sciences

T. Neupert *et al.*, arXiv:2102.04883

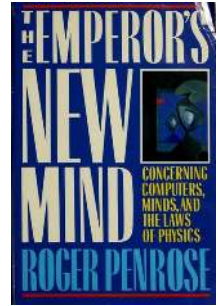
Literature: The Broader Context of AI



T. J. Sejnowski, *The Deep Learning Revolution*, MIT Press (2018)



E. J. Larson, *The Myth of Artificial Intelligence*, Harvard UP (2021)



R. Penrose, *The Emperor's New Mind*, Oxford UP (1989)

An enthusiast, a sceptic, and a physicist asking whether minds are computable (Gödel!) — read all three.

Prerequisites and Catching Up

You need:

- **linear algebra** — vectors, matrices, matrix products
- **calculus** — derivatives, gradients, chain rule
- **probability** — distributions, expectation values
- **Python & NumPy** — basics

You do *not* need:

- any prior machine learning
- any quantum physics

Catch-up resources:

- Prince, *Understanding Deep Learning*, Appendices (free)
- 3Blue1Brown, *Essence of Linear Algebra* & *Neural Networks* (videos)
- Deisenroth, Faisal, Ong, *Mathematics for Machine Learning* (free)

Unsure about something? Ask — there are no naive questions across disciplines.

Thank You!

Questions, project ideas, feedback: marcin.plodzien@uj.edu.pl

Website · GitHub · Google Scholar



